

WEEK 1-Practicals

Algorithm & Flowchart

1. Convert Fahrenheit to Centigrade

Algorithm: Fahrenheit to Centigrade Conversion

1. Start
2. Read temperature in Fahrenheit (F)
3. Calculate Centigrade using formula:

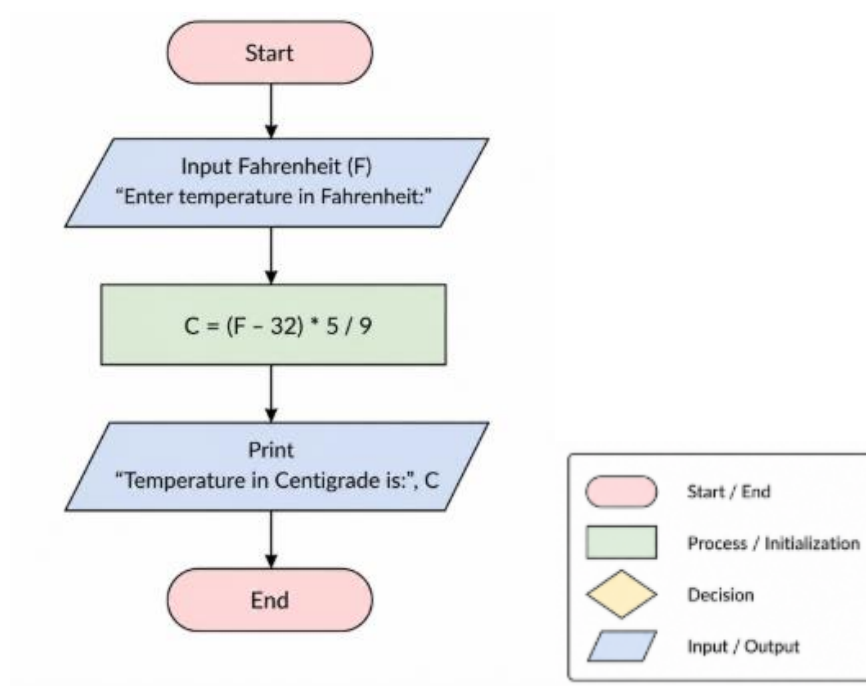
$$C = (F - 32) \times \frac{5}{9}$$

4. Display temperature in Centigrade
5. Stop

Python Code: Fahrenheit to Centigrade

```
# Program to convert Fahrenheit to Centigrade
# Input temperature in Fahrenheit
F = float(input("Enter temperature in Fahrenheit: "))
# Conversion formula
C = (F - 32) * 5 / 9
# Display result
print("Temperature in Centigrade is:", C)
```

Flow chart : Fahrenheit to Centigrade



2. Multiplication Table of 5 (Using Loop)

Algorithm:

1. **Start**
2. Initialize $n = 5$
3. Initialize $i = 1$
4. **While** $i \leq 10$
 - Calculate $\text{result} = n \times i$
 - Display $n \times i = \text{result}$
 - Increment $i = i + 1$
5. **Stop**

Python Code:

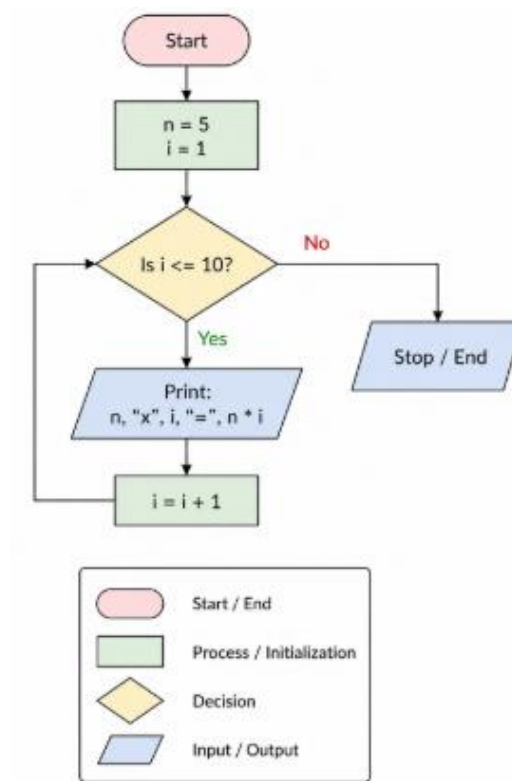
Program to display multiplication table of 5 using loop

$n = 5$

for i in range(1, 11):

$\text{print}(n, "x", i, "=", n * i)$

Flow chart:



3. Finding the Maximum of Three Forces

Algorithm:

Start

Read three forces: F_1, F_2, F_3

If $F_1 \geq F_2$ **and** $F_1 \geq F_3$

→ $\text{Max} = F_1$

Else if $F2 \geq F1$ and $F2 \geq F3$
→ Max = F2
Else
→ Max = F3
Display the maximum force
Stop

Python code :

```
# Program to find the maximum of three forces
# Input three forces
F1 = float(input("Enter Force 1: "))
F2 = float(input("Enter Force 2: "))
F3 = float(input("Enter Force 3: "))
# Decision making to find maximum force
if F1 >= F2 and F1 >= F3:
    max_force = F1
elif F2 >= F1 and F2 >= F3:
    max_force = F2
else:
    max_force = F3
# Display the result
print("The maximum force is:", max_force)
```

Flow chart:

